

Project Initialization and Planning Phase

Date	10 JUNE 2026
Team ID	
Project Title	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective is to revolutionize the educational evaluation process by implementing advanced natural language processing and audio analysis techniques, ensuring faster, highly objective, and accurate assessments of a student's conceptual understanding through spoken explanations..
Scope	The project comprehensively assesses and enhances the oral evaluation process, incorporating machine learning models (speech-to-text and semantic embedding networks) for a more robust, standardized, and efficient grading system.

Problem Statement		
Description	Addressing subjectiveness, evaluation delays, and inefficiencies in current oral examinations and conceptual understanding checks, which adversely affects grading consistency, operational turnaround times, and student feedback satisfaction..	
Impact	Solving these issues will result in improved academic evaluation efficiency, reduced grading biases, and an overall enhancement in the learning assessment pipeline, directly contributing to student progress tracking and institutional delivery success.	
Proposed Solution		
Approach	Employing machine learning techniques to analyze spoken audio files, convert them to text, and mathematically predict semantic alignment with master answer grids, creating a dynamic, adaptive, and highly objective voice-based evaluation framework.	
Key Features	- Implementation of an LLM/Transformer-based semantic assessment model (Sentence-BERT) coupled with automated speech-to-text processing (OpenAI Whisper). - Real-time acoustic processing for speech delivery flaw detection (identifying long silences, pause distributions, and filler word patterns using Librosa). - Continuous calibration capabilities allowing the evaluation system to seamlessly	
Resource Type	Description	Specification/Allocation
Hardware		

Computing Resources	CPU/GPU specifications, number of cores	NVIDIA T4 GPU or equivalent CUDA-enabled accelerator
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Resource Requirements

Memory	RAM specifications	16 GB
Storage	Disk space for data, models, and logs	50 GB SSD
Software		
Frameworks	Core UI application frameworks	Streamlit
Libraries	Additional computational and analysis packages	OpenAI-Whisper, sentencetransformers, librosa, reportlab, sounddevice
Development Environment	Integrated Development Environments	PyCharm, VS Code, Jupyter Notebook
Data		

Data	Academic concept master answers	Customizable CSV benchmark maps containing target terms and concept definitions
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